Acid Rain

Normal rain is slightly acidic (pH = 5.6) due to natural levels of CO_2, NO_2, and SO_2 in the air.

$$CO_{2(g)} + H_2O_{(l)} \longrightarrow H_2CO_{3(aq)}$$ dilute carbonic acid

Normal occurrences of CO_2:
- respiration
- forest fires
- volcanic erruptions

Nitrogen oxides occur naturally from:
- lightning strikes
- plant decay

$$2\ NO_{2(g)} + H_2O_{(l)} \longrightarrow HNO_{3(aq)} + HNO_{2(aq)}$$

nitric acid (strong) nitrous acid (weak)

Sulfur oxides occur naturally from:
- volcanic erruptions

1) $SO_{2(g)} + H_2O_{(l)} \longrightarrow H_2SO_{3(aq)}$ sulfurous acid (weak)

2) $SO_{2(g)} + O_{3(g)} \longrightarrow SO_{3(g)} + O_{2(g)}$

$SO_{3(g)} + H_2O_{(l)} \longrightarrow H_2SO_{4(aq)}$ sulfuric acid (strong)

Acid Precipitation
Any form of natural precipitation (rain, snow, etc.) that has an unusually high acidity (pH < 5.6)

Acid Deposition
- acid forming pollutants
- includes acid precipitation
- also includes dust and other dry particulate matter

Table 1 Effect of Water pH on Aquatic Life in Lakes and Rivers

pH	Effect
6.0	• Crustaceans, insects, and some plankton species begin to disappear.
5.0	• Changes occur in the plankton community. • Less desirable species of mosses and plankton appear. • Some fish populations die out. Smallmouth bass, walleye, brook trout, and salmon are the least tolerant of acidity. • Most fish eggs cannot hatch.
<5.0	• There are no fish. • The lake bottom is covered with undecayed material. • The surrounding shoreline may be dominated by mosses. • Terrestrial animals that depend on aquatic ecosystems are affected.

Table taken from Nelson Chemistry 12 College Preparation, p 326

Unit 4 Review

p487 #1-17, 19-24, 26, 27, 37, 29

p492 #1-21, 24, 25

$Ca(OH)_2$ HCl [0.20]